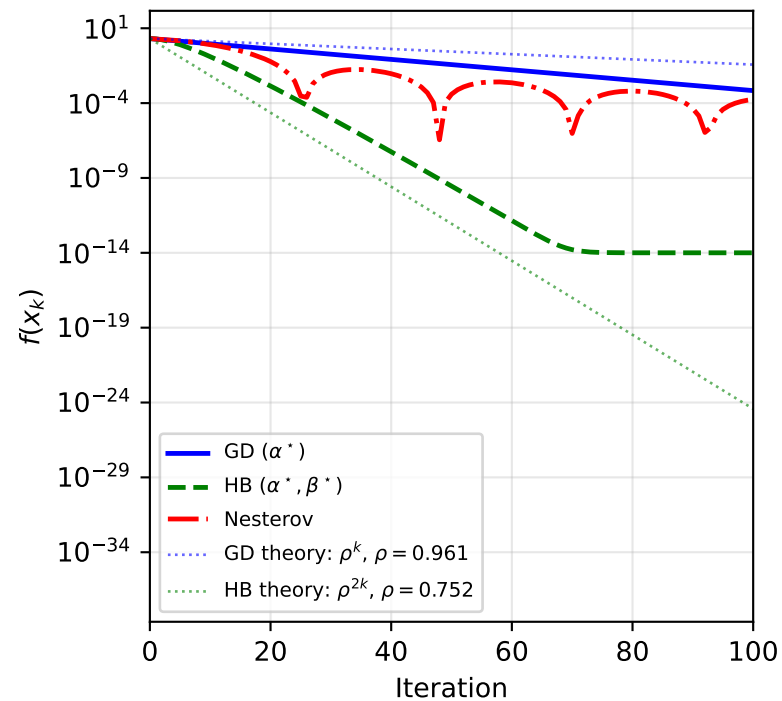
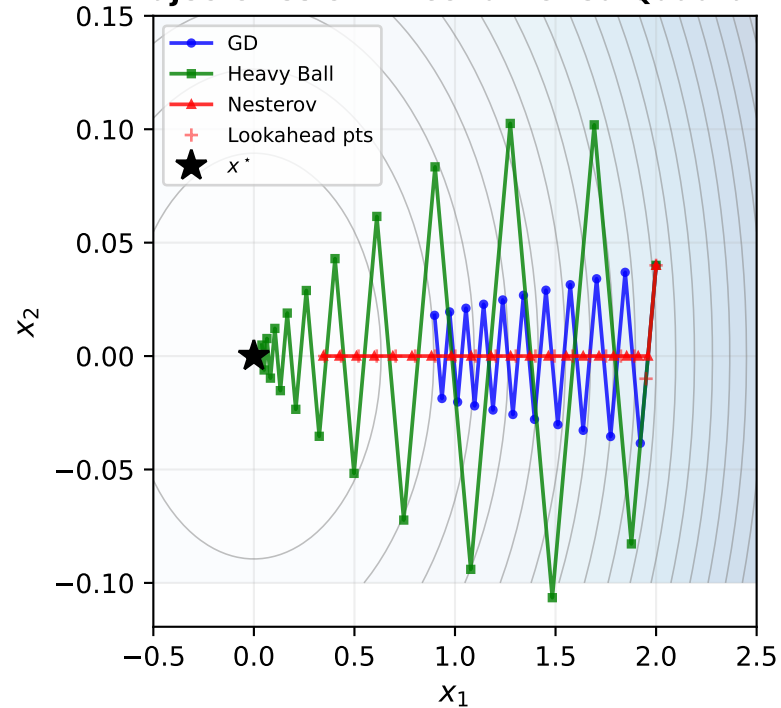


Nesterov vs Heavy Ball: Theory, Mechanisms, and Empirical Verification

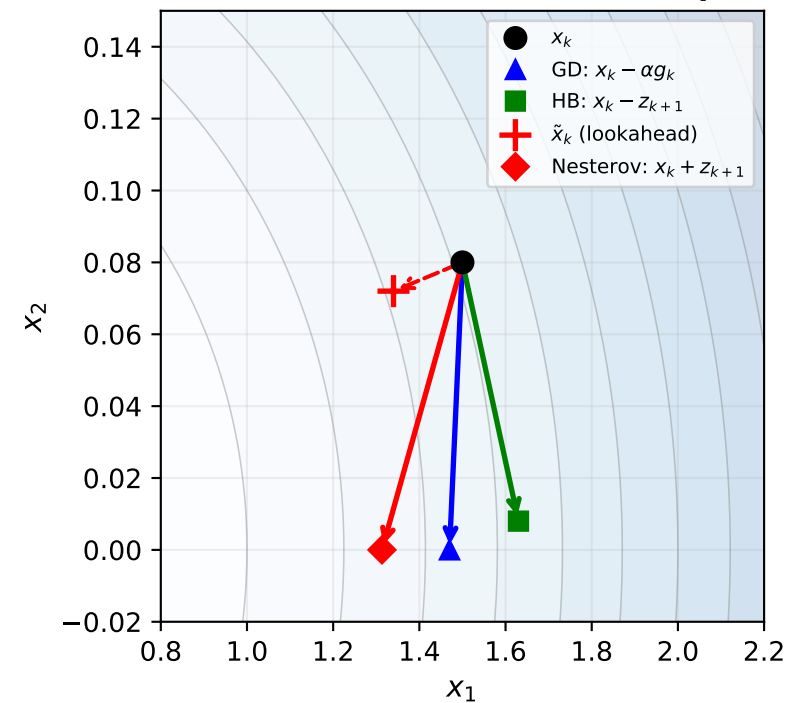
Quadratic ($\kappa = 50$): Theory vs Empirical



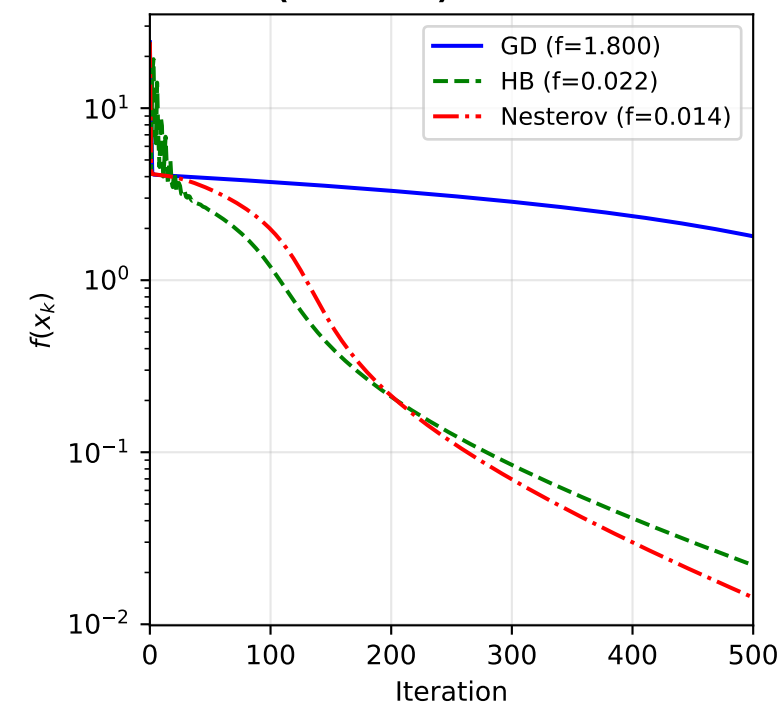
Trajectories on Ill-Conditioned Quadratic



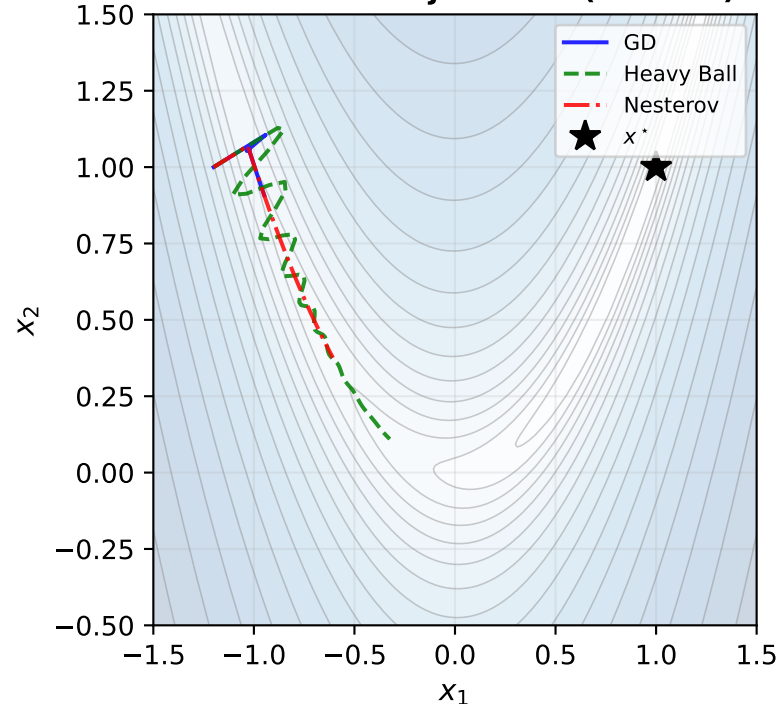
Nesterov Lookahead: One Step



Rosenbrock (500 iters): GD vs HB vs Nesterov



Rosenbrock Trajectories (80 iters)



Empirical vs Theoretical Iteration Counts

